

Commercial & Technical Proposal

Librika Smart Library & Learning Management Ecosystem (LMS/ILMS)

Executive Commercial Proposal

Client Information: Prepared for Indian Educational Institutions, Universities, and District Library Networks.

Project Overview: Librika bridges physical asset tracking (RFID/Barcodes) with digital reading gamification. It introduces high-speed AI cataloging, interactive subscription management, and auto-graded comprehension quizzes.

Key Objectives: Automate cataloging operations by 95% using vision models, increase student library utilization through B2B gamification, and establish a centralized peer repository for revision materials.

Project Deliverables Matrix

Deliverable Item	Specification Summary	Target Users
Unified Web Portal	Admin & Super Admin dashboards, Student Profile Hubs, Catalog OPAC	Super Admins, Librarians, Students
Mobile Applications	Native Android & iOS Apps compiled via Flutter wrapper	Students, Faculty Members
AI Scan Engine	Parallel vision cover scanner & double fallback crawler	Librarians / Catalog Administrators
AI Comprehension Portal	Nemotron comprehension quizzes with anti-cheat reading locks	Students / Evaluators
RFID Integrations	Interface modules connecting physical gates, readers, and tags	Librarians / Security Personnel

Chapter 2: Detailed Feature Breakdown by Module

Module Name	Functional Sub-Components & Core Deliverables
RFID Library Tracking	Real-time RFID gate triggers, batch tags writer, checkout reader sync, antenna controls.
Library System (ILMS)	Accession registers, circulation terms, overdue fines scheduler, barcode image generator.
LMS Engine	Course schedules, lecture slide attachments, homework submissions, gradebooks.
Mobile (Android/iOS)	Flutter wrappers, offline bookshelves, push notifications, scan-to-issue features.
AI Chatbot Widget	Virtual research assistant, book recommendations, `/api/chat` interactive assistant.
AI Semantic Search	Concept-based intent matching, `■ XX% Match` badges, hybrid offline fallback filters.
Recommendation Portal	Categorized content discovery, trending/new arrivals boards, category auto-filters.
Analytics Dashboards	Active checkouts stats, overdue registers, e-Library progress, student streaks graphs.
Moderation Panel	Notes & project upload reviews, review publication queues, compliance checklists.
Student & Faculty Portals	Personal bookshelves, reading streaks tracker, upload center, assignment grades review.

Chapter 3: Indian Cloud Sizing & AI Infrastructure Sizing

A. Cloud Infrastructure Sizing Options (Locally Hosted in India Data Centers)

Deployment Size	Sizing Specs & Components	DB & Backup Configuration	Est. Cost (INR/Mo)
Small (5 Schools)	4 vCPU, 16 GB RAM, 200 GB NVMe SSD	PostgreSQL, Daily S3 Backups	■6,000 – ■10,000
Medium (50 Schools)	8 vCPU, 32 GB RAM, 500 GB SSD, Load Balancer	PostgreSQL + Redis, Object Storage	■18,000 – ■35,000
Large (500+ Schools)	Kubernetes Cluster, Multi App Servers, DB Cluster	PostgreSQL Cluster + CDN, Daily Backups	■70,000 – ■2,00,000

B. AI Inference Infrastructure Comparison

Option	Hardware / Specs	Power & Electric Costs	Running Cost / Month
Option 1: API	OpenAI / OpenRouter Endpoint API. No GPUs required.	Zero server electrical overhead.	■5,000 – ■50,000 (Based on usage)
Option 2: Local (Small)	Self-Hosted: RTX 4090 / NVIDIA L40S (24GB VRAM). Run Llama 3.3 / Qwen / Gemma / Phi.	Power: ~450W per GPU. Est. Electricity: ■3,500/Mo.	Hardware: ■2.2L upfront. Maintenance: ■2,000/Mo.
Option 3: Local (Large)	Self-Hosted: Dedicated NVIDIA A100 / H100 (80GB VRAM) for large university clusters.	Power: ~700W per node. Est. Electricity: ■6,500/Mo.	Hardware: ■12L+ upfront. Maintenance: ■10,000/Mo.

Chapter 4: RFID Hardware & Vendor Pricing (India Market)

Recommended hardware partners: Zebra Technologies, Honeywell, Datalogic, Invengo, Times-7, GAO RFID, and local B2B vendors.

Hardware Item	Technical Specifications / Models	Estimated Unit Rate (INR)	Primary Purpose
RFID Book Tags	UHF EPC Gen2, ISO 18000-6C (e.g. Alien H3 / Monza R6)	₹8 – ₹15 / tag	Adhered inside book bindings for asset tracking.
RFID Reader	UHF Long-Range Fixed Reader (4-Port, Zebra FX9600 / Invengo)	₹75,000 – ₹1,20,000	Admin desk circulation and gate scanning controller.
RFID Security Gate	Dual-Aisle Pedestal Gates with integrated alarm lights	₹1,80,000 – ₹2,50,000	Antitheft security screening at entrance/exit.
Barcode Scanner	2D Handheld Scanner (Honeywell Voyager 1400g / Datalogic)	₹4,50,000 – ₹8,000	Quick backup catalog queries and user ID scans.
Label Printer	Thermal Transfer Barcode Printer (Zebra GT800 / TSC)	₹14,000 – ₹22,000	Prints barcode spine stickers and shelf identification labels.
Self Check Kiosk	Touchscreen kiosk with RFID Reader board, thermal receipt printer	₹1,50,000 – ₹2,50,000	Enables students to issue/return books independently.

Chapter 5: Resource Sizing, Budgets & Implementation Gantt

A. Software Development Team Resource Costs (Indian Market Rates)

Role Title	Monthly Rate (INR)	Allocation Duration	Est. Subtotal Cost (INR)
Project Manager	■1,20,000	6 Months	■7,20,000
Senior Backend Engineer	■1,80,000	6 Months	■10,80,000
Frontend UI Developer	■1,20,000	5 Months	■6,00,000
Flutter App Developer	■1,30,000	4 Months	■5,20,000
AI / VLM Engineer	■2,20,000	4 Months	■8,80,000
UI/UX Designer	■90,000	2 Months	■1,80,000
QA & Testing Engineer	■80,000	3 Months	■2,40,000
DevOps Cloud Engineer	■1,50,000	2 Months	■3,00,000
Total Resources Sizing	-	-	■45,20,000

B. Project Implementation Timeline (6-Month Gantt Sequence)

Phase	Key Milestone Deliverables	Timeline Sequence	Primary Dependencies
1. Planning	Requirements definition, DB schema alignment	Month 1 (Weeks 1-2)	Stakeholder sign-off
2. UI/UX Design	Wireframes, mobile mockups, billing interface designs	Month 1 (Weeks 3-4)	Phase 1 specs
3. Backend Dev	Core Flask API, database setup, WAL configurations	Months 2-3	Database isolation schemas
4. Frontend Dev	Student/Admin web dashboards, Flutter mobile wrappers	Months 3-4	Backend API schemas
5. AI Engineering	Nemotron scanner integration, AI quiz locks, semantic search	Month 4	VLM model training API keys
6. System Testing	Concurrency tests, RFID hardware integration checks	Month 5	Module completions
7. Deployment & Training	Server setup, staging launch, librarian training runs	Month 6	User acceptance validation

Chapter 6: Final B2B Commercial Quotations & AMC Plans

A. Commercial Negotiation Packages (Indian Client Tiers)

Package Level	Target Audience Profile	Included Scope Items	Price Package Estimate
Basic Tier	Small schools, local public libraries	Web Portal, OPAC, Barcode cataloging, standard search.	■25 – ■35 Lakhs
Professional Tier	Standard colleges, multi-school campuses	Web portal, Android/iOS Apps, AI scanner + semantic search, billing.	■45 – ■65 Lakhs
Enterprise Tier	University systems, state library groups	Full RFID integration, AI comprehension quizzes, custom state dashboards.	■80 Lakhs – ■1.5 Crore+

B. Annual Maintenance Contract (AMC) Tiers

AMC Level	Contract Fee	Included Support SLA Items
Basic Support	15% of License/Yr	Standard bug fixing, monthly database backups, email-only support tickets.
Premium Support	20% of License/Yr	Weekly backups, minor feature updates, server patch monitoring, 24-hr SLA response.
Enterprise Support	25% of License/Yr	Real-time replica backups, uptime guarantees (99.9%), security audits, on-site hardware triage.

C. Optional Enterprise Add-Ons

- **Aadhaar / SSO Integration:** ■1,80,000 upfront. (Secure student identity check).
- **DigiLocker Integration:** ■1,50,000 upfront. (Auto-saving study certificates).
- **WhatsApp & SMS Gateway API:** ■90,000 upfront. (Auto-notifications on due dates).
- **UPI Payment Gateway:** ■1,20,000 upfront. (Auto-paying overdue fines).

Proposal Authorized By: Librika Commercial Sales Team.

*Pricing excludes standard GST (18%) where applicable.